

OpenSlot Milestone 02 - Database Design Package

A) Relational / ER Diagram (text form)

[USERS] (user_id PK)

- name
- email UNIQUE
- role CHECK(student|professor)
- created_at

[COURSES] (course_id PK)

- course_code
- course_name
- term

[OFFICE_HOUR_SLOTS] (slot_id PK)

- professor_id FK -> USERS.user_id
- course_id FK -> COURSES.course_id
- start_time
- end_time
- mode CHECK(in_person|virtual)
- location_or_link
- visibility CHECK(public|private)
- status CHECK(draft|posted|cancelled)
- booked_by FK -> USERS.user_id (nullable)
- notes
- created_at
- updated_at

[APPOINTMENTS] (appointment_id PK)

- slot_id FK -> OFFICE_HOUR_SLOTS.slot_id UNIQUE
- student_id FK -> USERS.user_id
- status CHECK(booked|cancelled|completed)
- notes
- created_at

Cardinality summary:

- USERS (professors) 1 -> many OFFICE_HOUR_SLOTS
- COURSES 1 -> many OFFICE_HOUR_SLOTS
- USERS (students) 1 -> many APPOINTMENTS
- OFFICE_HOUR_SLOTS 1 -> zero/one APPOINTMENTS

B) SQL CREATE TABLE Statements

-- OpenSlot relational schema (Milestone 02)

```
CREATE TABLE IF NOT EXISTS users (  
  user_id INTEGER PRIMARY KEY,  
  name VARCHAR(120) NOT NULL,  
  email VARCHAR(255) NOT NULL UNIQUE,  
  role VARCHAR(20) NOT NULL CHECK (role IN ('student', 'professor')),  
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP  
);
```

```
CREATE TABLE IF NOT EXISTS courses (  
  course_id INTEGER PRIMARY KEY,  
  course_code VARCHAR(20) NOT NULL,  
  course_name VARCHAR(160) NOT NULL,  
  term VARCHAR(40) NOT NULL,  
  UNIQUE (course_code, term)  
);
```

```

CREATE TABLE IF NOT EXISTS office_hour_slots (
    slot_id INTEGER PRIMARY KEY,
    professor_id INTEGER NOT NULL,
    course_id INTEGER NOT NULL,
    start_time TIMESTAMP NOT NULL,
    end_time TIMESTAMP NOT NULL,
    mode VARCHAR(20) NOT NULL CHECK (mode IN ('in-person', 'virtual')),
    location_or_link VARCHAR(300),
    visibility VARCHAR(20) NOT NULL DEFAULT 'public' CHECK (visibility IN ('public', 'private')),
    status VARCHAR(20) NOT NULL DEFAULT 'draft' CHECK (status IN ('draft', 'posted', 'cancelled')),
    booked_by INTEGER,
    notes TEXT,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (professor_id) REFERENCES users (user_id),
    FOREIGN KEY (course_id) REFERENCES courses (course_id),
    FOREIGN KEY (booked_by) REFERENCES users (user_id),
    CHECK (end_time > start_time)
);

```

```

CREATE TABLE IF NOT EXISTS appointments (
    appointment_id INTEGER PRIMARY KEY,
    slot_id INTEGER NOT NULL UNIQUE,
    student_id INTEGER NOT NULL,
    status VARCHAR(20) NOT NULL DEFAULT 'booked' CHECK (status IN ('booked', 'cancelled', 'completed')),
    notes TEXT,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (slot_id) REFERENCES office_hour_slots (slot_id),
    FOREIGN KEY (student_id) REFERENCES users (user_id)
);

```

```

CREATE INDEX IF NOT EXISTS idx_slots_professor_time
ON office_hour_slots (professor_id, start_time);

```

```

CREATE INDEX IF NOT EXISTS idx_slots_course_time
ON office_hour_slots (course_id, start_time);

```

```

CREATE INDEX IF NOT EXISTS idx_appointments_student
ON appointments (student_id);

```

C) Normalization notes

- 1NF: atomic attributes only.
- 2NF: every non-key attribute depends on its full key.
- 3NF: no transitive dependency among non-key attributes.